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Inventor(s)	Iwaki; Noriaki

Substrate working machine

Abstract

Provided is a substrate working machine including a holding device configured to hold a substrate in which multiple through-holes are formed, an inserting device configured to insert multiple terminals of a component into the multiple through-holes of the substrate held by the holding device, and an imaging device configured to simultaneously image a shape of a pair of pins and a pair of through-holes of the multiple through-holes, in which positions of the pair of through-holes and the shape of the pair of pins are calculated based on image data captured by the imaging device.

Inventors:	Iwaki; Noriaki (Hekinan, JP)
Applicant:	FUJI CORPORATION (Chiryu, JP)
Family ID:	77771956
Assignee:	FUJI CORPORATION (Chiryu, JP)
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Primary Examiner: Worden; Thomas E
Assistant Examiner: Cameron; Atticus A
Attorney, Agent or Firm: Oblon, McClelland, Maier & Neustadt, L.L.P.

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a substrate working machine for inserting multiple terminals of a component into multiple through-holes formed in a substrate.

BACKGROUND ART

(2) In the following Patent Literature, a substrate working machine for inserting multiple terminals of a component into multiple through-holes formed in a substrate is described.

PATENT LITERATURE

(3) Patent Literature 1: JP-A-H1-238197

BRIEF SUMMARY

Technical Problem

(4) An object of the present specification is to insert multiple terminals of a component into multiple through-holes formed in a substrate.

Solution to Problem

(5) In order to solve the above-mentioned problems, the present specification discloses a substrate working machine including a holding device configured to hold a substrate in which multiple through-holes are formed, an inserting device configured to insert multiple terminals of a component into the multiple through-holes of the substrate held by the holding device, and an imaging device configured to simultaneously image a shape of a pair of pins and a pair of through-holes of the multiple through-holes, in which positions of the pair of through-holes and the shape of the pair of pins are calculated based on image data captured by the imaging device.

Advantageous Effects

(6) In the case of the substrate working machine of the present disclosure, the shape of the pair of pins and the pair of through-holes are simultaneously imaged and the positions of the pair of through-holes and the shape of the pair of pins are calculated based on the captured image data thereof. As a result, for example, it is possible to appropriately insert the multiple terminals into the multiple through-holes by calculating the positions of the pair of through-holes and the shape of the pair of pins based on the positions of the pair of through-holes and the shape of the pair of pins.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view showing a component mounter.
- (2) FIG. 2 is a perspective view showing a component mounting device.
- (3) FIG. 3 is a view showing a component holding tool.
- (4) FIG. 4 is a perspective view showing a cut-and-clinch unit and a unit moving device.
- (5) FIG. 5 is a perspective view showing the cut-and-clinch unit.
- (6) FIG. 6 is a schematic view showing the cut-and-clinch unit during the cutting and bending of leads of a lead component.
- (7) FIG. 7 is a block diagram showing a control device.
- (8) FIG. 8 is a schematic view showing a state where a through-hole with pins inserted therein is imaged by an imaging device.
- (9) FIG. 9 is a captured image of a through-hole with pins inserted therein.
- (10) FIG. 10 is a captured image of a through-hole with pins inserted therein.
- (11) FIG. 11 is a captured image of a through-hole with pins inserted therein.
- (12) FIG. 12 is a perspective view showing a cut-and-clinch unit.
- (13) FIG. 13 is a perspective view showing the cut-and-clinch unit.
- (14) FIG. 14 is a schematic view showing a state where leads of a lead component are inserted into through-holes of a circuit base material by means of the cut-and-clinch unit.
- (15) FIG. 15 is a schematic view showing a state where a through-hole with guide pins inserted therein is imaged by the imaging device.

DESCRIPTION OF EMBODIMENTS

- (16) Hereinafter, exemplary embodiments of the present disclosure will be described in detail with reference to drawings.
- (17) FIG. 1 shows component mounter 10. Component mounter 10 is a device that performs a component mounting operation on circuit base material 12. Component mounter 10 includes device main body 20, base material conveyance and holding device 22, component mounting device 24, mark camera 26, part camera 28, component supply device 30, bulk component supply device 32, cut-and-clinch unit (refer to FIG. 5) 34, unit moving device (refer to FIG. 4) 36, and control device (refer to FIG. 7) 38. Examples of circuit base material 12 include a circuit substrate, a substrate

having a three-dimensional structure, and the like, and examples of a circuit substrate include a printed wiring board, a printed circuit board, and the like.

(18) Device main body **20** is composed of frame **40** and beam **42** suspended on frame **40**. Base material conveyance and holding device **22** is disposed at the center of frame **40** in a front-rear direction, and has conveyance device **50** and clamp device **52**. Conveyance device **50** is a device for conveying circuit base material **12**, and clamp device **52** is a device for holding circuit base material **12**. As a result, base material conveyance and holding device **22** conveys circuit base material **12** and fixedly holds circuit base material **12** at a predetermined position. In the description below, the conveyance direction of circuit base material **12** will be referred to as an X-direction, a horizontal direction perpendicular to the X-direction will be referred to as a Y-direction, and a vertical direction will be referred to as a Z-direction. That is, the width direction of component mounter **10** is the X-direction, and the front-rear direction is the Y-direction.

(19) Component mounting device **24** is disposed on beam **42** and includes two work heads **60** and **62** and work head moving device **64**. Work head moving device **64** includes X-direction moving device **68**, Y-direction moving device **70**, and Z-direction moving devices **72**. In addition, two work heads **60** and **62** are integrally moved to any position on frame **40** by X-direction moving device **68** and Y-direction moving device **70**. In addition, each of work heads **60** and **62** is positioned and mounted on sliders **74** and **76** such that each of work heads **60** and **62** can be detached and attached by a worker without using any tool, and Z-direction moving devices **72** individually move sliders **74** and **76** in an up-down direction. That is, work heads **60** and **62** are moved in the up-down direction individually by Z-direction moving devices **72**.

(20) In addition, as illustrated in FIG. 2, component holding tool **77** is attached to a lower end surface of each of work heads **60** and **62**. Component holding tool **77** is a so-called chuck, and includes main body **78** and pair of claws **79** as illustrated in FIG. 3. Pair of claws **79** is disposed to extend downwards from a lower surface of main body **78** and linearly slides to move to and away from each other. As a result, component holding tool **77** holds a component with pair of claws **79** by causing pair of claws **79** to move to each other, and releases the component from between pair of claws **79** by causing pair of claws **79** to move away from each other. In addition, each of work heads **60** and **62** is provided with a rotation device (not illustrated) that rotates component holding tool **77** around the vertical axis, so that the posture of a component held by component holding tool **77** can be changed by means of the operation of the rotation device.

(21) Mark camera **26** is attached to slider **74** in a state of facing downward on the vertical line as illustrated in FIG. 2, and mark camera **26** moves together with work head **60** in the X-direction, the Y-direction, and the Z-direction. As a result, mark camera **26** images any position on frame **40**. As illustrated in FIG. 1, part camera **28** is disposed between base material conveyance and holding device **22** on frame **40** and component supply device **30** in a state of facing upward on the vertical line. As a result, part camera **28** images components held by component holding tools **77** of work heads **60** and **62**.

(22) Component supply device **30** is disposed at a first end portion of frame **40** in the front-rear direction. Component supply device **30** includes tray-type component supply device **86** and feeder-type component supply device (refer to FIG. 6) **88**. Tray-type component supply device **86** is a device that supplies a component in a state in which the component is placed on a tray. Feeder-type component supply device **88** is a device that supplies a component by using a tape feeder or a stick feeder (not illustrated).

(23) Bulk component supply device **32** is disposed at a second end portion of frame **40** in the front-rear direction. Bulk component supply device **32** is a device that aligns multiple components in a scattered state and supplies the components in an aligned state. That is, bulk component supply device **32** is a device that aligns multiple components in any posture to a predetermined posture, and supplies the components of the predetermined posture.

(24) Examples of components which are supplied by component supply device **30** and bulk

component supply device **32** include electronic circuit components, constituent components of a solar cell, constituent components of a power module, and the like. In addition, the electronic circuit components include a component with a lead, a component without a lead, and the like.

(25) Cut-and-clinch unit **34** and unit moving device **36** are disposed below conveyance device **50**. Cut-and-clinch unit **34** can be moved to any position below conveyance device **50** by unit moving device **36**. Cut-and-clinch unit **34** is a device for cutting and bending leads (refer to FIG. **6**) **108** of lead component (refer to FIG. **6**) **106** that are inserted into through-holes **104** formed in circuit base material **12** (refer to FIG. **6**). As illustrated in FIG. **5**, cut-and-clinch unit **34** includes pair of sliding bodies **112**. Pair of sliding bodies **112** are slidably supported by slide rail **116** disposed to extend in the X-direction. With this configuration, pair of sliding bodies **112** moves to and away from each other in the X-direction. In addition, the distance between pair of sliding bodies **112** is controllably changed by the driving of electromagnetic motor (refer to FIG. **7**) **118**.

(26) In addition, each of pair of sliding bodies **112** includes fixed portion **120** and movable portion **122** and is slidably held by slide rail **116** at fixed portion **120**. In addition, movable portion **122** is held by fixed portion **120** to be slidable in the X-direction. In addition, movable portion **122** is controllably slid in the X-direction with respect to fixed portion **120** by the driving of electromagnetic motor (refer to FIG. **7**) **128**.

(27) In addition, as illustrated in FIG. **6**, an upper end portion of fixed portion **120** has a tapered shape, and first insertion hole **130** is formed to penetrate the upper end portion in the up-down direction. In addition, an opening edge of first insertion hole **130** at an upper end surface is fixed blade **131**. Meanwhile, an upper end portion of movable portion **122** also has a tapered shape, and bent portion **133**, which is bent into an L-shape, is formed at the upper end portion. Bent portion **133** extends above an upper end surface of fixed portion **120**. In addition, first insertion hole **130**, which is open at the upper end surface of fixed portion **120**, is covered by bent portion **133**, and second insertion hole **136** is formed in bent portion **133** such that second insertion hole **136** faces first insertion hole **130**. Note that an opening edge of second insertion hole **136** at a lower end surface of bent portion **133** is movable blade **138**.

(28) In addition, unit moving device **36**, as illustrated in FIG. **4**, includes X-direction moving device **150**, Y-direction moving device **152**, Z-direction moving device **154**, and rotation device **156**. X-direction moving device **150** includes slide rails **160** and X-slider **162**. Slide rails **160** are disposed to extend in the X-direction, and X-slider **162** is slidably held by slide rails **160**. In addition, X-slider **162** moves in the X-direction by the driving of electromagnetic motor (refer to FIG. **7**) **164**. Y-direction moving device **152** includes slide rails **166** and Y-slider **168**. Slide rails **166** are disposed on X-slider **162** to extend in the Y-direction, and Y-slider **168** is slidably held by slide rails **166**. In addition, Y-slider **168** moves in the Y-direction by the driving of electromagnetic motor (refer to FIG. **7**) **170**. Z-direction moving device **154** includes slide rails **172** and Z-slider **174**. Slide rails **172** are disposed on Y-slider **168** to extend in the Z-direction, and Z-slider **174** is slidably held by slide rails **172**. In addition, Z-slider **174** moves in the Z-direction by the driving of electromagnetic motor (refer to FIG. **7**) **176**.

(29) In addition, rotation device **156** includes rotary table **178** having a substantially circular disc-like shape. Rotary table **178** is supported on Z-slider **174** such that rotary table **178** can rotate around a vertical axis thereof and rotates by the driving of electromagnetic motor (refer to FIG. **7**) **180**. In addition, on rotary table **178**, cut-and-clinch unit **34** is positioned and disposed such that cut-and-clinch unit **34** can be detached and attached through a one-touch action performed by a worker without using any tool. With such a structure, cut-and-clinch unit **34** is moved to any position between a pair of conveyance lanes of base material conveyance and holding device **22**, which is for conveyance of circuit base material **12**, by X-direction moving device **150**, Y-direction moving device **152**, and Z-direction moving device **154** and is rotated to any angle by rotation device **156**. As a result, cut-and-clinch unit **34** can be positioned at any position below circuit base material **12** held by clamp device **52**.

(30) As illustrated in FIG. 7, control device **38** includes controller **190**, multiple drive circuits **192**, and image processing device **196**. Multiple drive circuits **192** are connected to conveyance device **50**, clamp device **52**, work heads **60** and **62**, work head moving device **64**, tray-type component supply device **86**, feeder-type component supply device **88**, bulk component supply device **32**, and electromagnetic motors **118**, **128**, **164**, **170**, **176**, and **180**, which are described above. Controller **190** includes a CPU, a ROM, a RAM, and the like, is mainly composed of a computer, and is connected to multiple drive circuits **192**. As a result, the operations of base material conveyance and holding device **22**, component mounting device **24**, and the like are controlled by controller **190**. Furthermore, controller **190** is also connected to image processing device **196**. Image processing device **196** is a device that processes image data obtained by mark camera **26** and part camera **28**, and controller **190** acquires various pieces of information from the image data.

(31) With the configuration as described above, component mounter **10** performs a component mounting operation on circuit base material **12** held by base material conveyance and holding device **22**. Although component mounter **10** can mount various types of components on circuit base material **12**, a case where lead component **106** is mounted on circuit base material **12** will be described below.

(32) Specifically, circuit base material **12** is conveyed to a work position by conveyance device **50** of base material conveyance and holding device **22** and is fixedly held by clamp device **52** at the position. Next, mark camera **26** moves to a position above circuit base material **12** and images circuit base material **12**. As a result, information on the holding position of circuit base material **12** and the like is obtained. In addition, component supply device **30** or bulk component supply device **32** supplies lead component **106** to work heads **60** and **62** at a predetermined supply position. Then, any of work heads **60** and **62** moves to a position above a component supply position and component main body (refer to FIG. 6) **200** of lead component **106** is gripped by pair of claws **79** of component holding tool **77**.

(33) Subsequently, work heads **60** and **62** holding the lead component **106** move to positions above part camera **28**, and part camera **28** images lead component **106** held by pair of claws **79** of component holding tool **77**. As a result, information on the holding position of the component and the like is obtained. Subsequently, work heads **60** and **62** holding lead component **106** move to positions above circuit base material **12** and errors of the holding position of circuit base material **12**, errors of the holding position of the component, and the like are corrected. Then, leads **108** of lead component **106** held by component holding tool **77** are inserted into through-holes **104** formed in circuit base material **12**. At this time, cut-and-clinch unit **34** is moved to a position below circuit base material **12**.

(34) Specifically, in cut-and-clinch unit **34**, the distance between pair of sliding bodies **112** is moved and positioned by the operation of electromagnetic motor **118** such that the distance between second insertion holes **136** of pair of sliding bodies **112** is the same as the distance between two through-holes **104** formed in circuit base material **12**. In addition, rotation device **156** is operated and positioned such that the direction of alignment of two through-holes **104** of circuit base material **12** coincides with the direction of alignment of two second insertion holes **136** of pair of sliding bodies **112**.

(35) Then, X-direction moving device **150** and Y-direction moving device **152** are operated and cut-and-clinch unit **34** is moved and positioned such that the XY-direction coordinates of second insertion holes **136** coincide with the XY-direction coordinates of through-holes **104** of circuit base material **12**. As a result, cut-and-clinch unit **34** is moved along the XY-directions and second insertion holes **136** of sliding bodies **112** and through-holes **104** of circuit base material **12** are positioned in a state of overlapping with each other in the up-down direction.

(36) Next, cut-and-clinch unit **34** is raised and positioned by the operation of Z-direction moving device **154** such that an upper surface of movable portion **122** comes into contact with a lower surface of circuit base material **12** or is positioned slightly lower than the lower surface of circuit

base material **12**. As described above, the operations of X-direction moving device **150**, Y-direction moving device **152**, Z-direction moving device **154**, and rotation device **156** are controlled such that cut-and-clinch unit **34** is positioned below circuit base material **12** in a state where second insertion holes **136** of sliding bodies **112** and through-holes **104** of circuit base material **12** overlap with each other.

(37) In addition, in a case where leads **108** of lead component **106** held by component holding tool **77** are inserted into through-holes **104** of circuit base material **12**, distal end portions of leads **108** are inserted into first insertion holes **130** of fixed portions **120** via second insertion holes **136** of movable portions **122** of cut-and-clinch unit **34**, as illustrated in FIG. **6**. Next, in a case where the distal end portions of leads **108** are inserted into first insertion holes **130** of fixed portions **120**, movable portions **122** of pair of sliding bodies **112** slide away from each other by the operation of electromagnetic motor **128**. As a result, leads **108** are cut by fixed blades **131** of first insertion holes **130** and movable blades **138** of second insertion holes **136**.

(38) In addition, pair of movable portions **122** further slides away from each other even after leads **108** are cut. Therefore, new distal end portions of leads **108** due to the cutting are bent as movable portions **122** slide. As a result, pair of leads **108** is bent away from each other, so that lead component **106** is mounted on circuit base material **12** in a state where leads **108** are prevented from coming out of through-holes **104**.

(39) As described above, in component mounter **10**, leads **108** of lead component **106** are inserted into through-holes **104** of circuit base material **12**, and distal ends of leads **108** are cut and bent by cut-and-clinch unit **34**. Therefore, although it is desired to suitably insert leads **108** into through-holes **104** of circuit base material **12**, a method of improving the insertion accuracy has not been established in a conventional component mounter. Therefore, in the case of component mounter **10**, a jig provided on which pins are formed is used and the pins of the jig held by work heads **60** and **62** are inserted into the through-holes of circuit base material **12** so that the accuracy of insertion of the leads of the lead component into the through-holes of the circuit base material is improved based on imaging data of the through-holes obtained in a state where the pins are inserted.

(40) Specifically, as illustrated in FIG. **8**, jig **200** is composed of component main body **202** and pair of pins **204**. Component main body **202** is substantially block-shaped, and pair of pins **204** extends linearly and vertically from the bottom surface of component main body **202** in the same direction. Pair of pins **204** is standardized, and the pitch, the outer diameter, the length dimension, and the like of the pins **204** are set in advance. In other words, standardized pair of pins **204** extends in the vertical direction from a surface of component main body **202** with high accuracy and the accuracy is secured. In addition, component main body **202** of jig **200** is clamped by pair of claws **79** of component holding tool **77**, so that jig **200** is held by component holding tool **77**.

(41) In addition, substrate **220** serving as a jig used for jig **200** is also prepared. As illustrated in FIG. **9**, four through-holes **222** are formed in substrate **220**, and four through-holes **222** are arranged in two rows and two columns. Four through-holes **222** are also standardized, and the pitch of two of four through-holes **222** that are adjacent to each other, the inner diameters of the through-holes **222**, and the like are accurately processed to be dimensions set in advance. Note that the pitch of two of four through-holes **222** that are adjacent to each other and the pitch of pair of pins **204** of jig **200** are processed to be the same as each other. Incidentally, the pitch of two through-holes **222** is the distance between the center of first through-hole **222** and the center of second through-hole **222**, and the pitch of pair of pins **204** is the distance between the center of an end surface of first pin **204** and the center of an end surface of second pin **204**. In addition, the inner diameter of through-hole **222** is approximately three times the outer diameter of pin **204**.

(42) Substrate **220** having such a structure is conveyed to a work position by conveyance device **50** and is clamped at the work position by clamp device **52**. Then, substrate **220** positioned and fixed is imaged by mark camera **26**, and the position of each through-hole **222** is calculated based on the imaging data thereof. In addition, jig **200** held by component holding tool **77** is imaged by part

camera **28**, and the position of the pair of pins is calculated based on the imaging data thereof. Then, work heads **60** and **62** are moved by the operation of work head moving device **64** such that pair of pins **204** coincides with two of four through-holes **222** in the up-down direction, that is, the XY-coordinates of pair of pins **204** coincide with the XY-coordinates of two through-holes **222**. Subsequently, work heads **60** and **62** are lowered by the operation of work head moving device **64**, so that pair of pins **204** is inserted into two through-holes **222** as illustrated in FIG. **8**. At this time, work heads **60** and **62** are lowered such that end surfaces of pins **204** and a lower surface of substrate **220** coincide with each other in the up-down direction, that is, the end surfaces of pins **204** and the lower surface of substrate **220** are flush with each other.

(43) In addition, on a lower surface side of substrate **220**, one imaging device **230** is attached to rotary table **178** of unit moving device **36** instead of cut-and-clinch unit **34** attached thereto. As described above, the cut-and-clinch unit **34** can be positioned, detached from, and attached to rotary table **178** through a one-touch action performed by a worker without using any tool and imaging device **230** is also attachably and detachably attached to rotary table **178**, from which cut-and-clinch unit **34** has been removed, in a state of being positioned while facing upward on the vertical line. Then, imaging device **230** attached to rotary table **178** is moved to a position below through-holes **222** of substrate **220** by the operation of unit moving device **36**.

(44) Therefore, pair of pins **204** inserted into two through-holes **222** of substrate **220** is imaged by imaging device **230**. That is, pair of pins **204** and two through-holes **222** with pair of pins **204** inserted therinto are simultaneously imaged by imaging device **230**. Note that imaging device **230** is a two-dimensional imaging device including an imaging element in which cells are two-dimensionally arranged and images a three-dimensional object to obtain a two-dimensional image. Therefore, the depth of field of imaging device **230** is relatively narrow. Therefore, as described above, lower end surfaces of pins **204** inserted into through-holes **222** and the lower surface of substrate **220** are made flush with each other and are positioned within the depth of field of the imaging device, so that an appropriate captured image of the lower end surfaces of pins **204** and through-holes **222** is acquired.

(45) In a case where standardized pins **204** are inserted into standardized through-holes **222** in this manner, an ideal insertion state is a state where pair of pins **204** is inserted into the centers of two through-holes **222**. That is, as described above, in a case where two through-holes **222** with pair of pins **204** inserted therinto are imaged by one imaging device **230**, ideally, the centers of pair of pins **204** and the centers of two through-holes **222** coincide with each other without being offset from each other in the imaging data thereof, as illustrated in FIG. **9**. However, as illustrated in FIG. **10**, there is a case where the centers of pair of pins **204** and the centers of two through-holes **222** are offset from each other because of a slight difference in product accuracy between component mounters, a decrease in insertion accuracy with time caused by use of component mounter **10**, or the like.

(46) Therefore, for each component mounter, the control device calculates the position of a pair of through-holes and the shape of a pair of pins based on the position of the pair of through-holes and the shape of the pair of pins, acquires a correction value for appropriately inserting terminals of a component held by a work head into multiple through-holes of a circuit base material, and can appropriately insert the multiple terminals into the multiple through-holes based on the acquired correction value. Therefore, in component mounter **10**, the amount of deviation between inserted pins **204** and through-holes **222** is calculated based on imaging data of two through-holes **222** with pair of pins **204** inserted therinto, and a correction value at the time of inserting the pair of pins into the pair of through-holes is acquired based on the calculated amount of deviation. That is, for example, in a case where the centers of pair of pins **204a** and **204b** and the centers of two through-holes **222a** and **222b** are offset from each other as illustrated in FIG. **10**, the amount of deviation ($\Delta X1$, $\Delta Y1$) in the XY-directions between the center of first pin **204a** and the center of first through-hole **222a** and the amount of deviation ($\Delta X2$, $\Delta Y2$) in the XY-directions between the

center of second pin **204b** and the center of second through-hole **222b** are calculated. In addition, the angle of deviation (**81**) formed by a straight line connecting the center of first pin **204a** and the center of second pin **204b** and a straight line connecting the center of first through-hole **222a** and the center of second through-hole **222b** is also calculated.

(47) Subsequently, through-holes **222** into which pair of pins **204** is inserted are changed and other pair of through-holes **222** is imaged. For example, component holding tool **77** holding jig **200** is rotated by 90 degrees by the rotation device. Then, according to the above-described procedure, pair of pins **204** is inserted into two through-holes **222**. In the case of component holding tool **77** rotated by 90 degrees by the rotation device, since pair of pins **204** of held jig **200** also rotates by 90 degrees, as illustrated in FIG. **11**, first pin **204a** is inserted into first through-hole **222a** and second pin **204b** is inserted into second through-hole **222c**. Then, pair of pins **204a** and **204b** and two through-holes **222a** and **222c** with the pair of pins inserted thereinto are simultaneously imaged by one imaging device **230**, and the amount of deviation between pins **204** and through-holes **222** is calculated. That is, the amount of deviation ($\Delta X3$, $\Delta Y3$) in the XY-directions between the center of first pin **204a** and the center of first through-hole **222a** and the amount of deviation ($\Delta X4$, $\Delta Y4$) in the XY-directions between the center of second pin **204b** and the center of second through-hole **222c** are calculated. In addition, the angle of deviation ($\theta2$) formed by a straight line connecting the center of pair of pins **204a** and the center of pin **204b** and a straight line connecting the center of pair of through-holes **222a** and the center of through-hole **222c** is also calculated.

(48) Furthermore, through-holes **222** into which pair of pins **204** is inserted are changed and other pair of through-holes **222** is imaged. For example, component holding tool **77** holding jig **200** is further rotated by 90 degrees by a rotation device. Then, according to the above-described procedure, pair of pins **204** is inserted into two through-holes **222**. In the case of component holding tool **77** further rotated by 90 degrees by the rotation device, since pair of pins **204** of held jig **200** also rotates by 90 degrees, although not shown, first pin **204a** is inserted into first through-hole **222b** and second pin **204b** is inserted into second through-hole **222a**. Then, pair of pins **204a** and **204b** and two through-holes **222a** and **222c** with the pair of pins inserted thereinto are simultaneously imaged from below by one imaging device **230**, and the amount of deviation between pins **204** and through-holes **222** is calculated. That is, the amount of deviation in the XY-directions between the center of pin **204a** and the center of through-hole **222b** and the amount of deviation in the XY-directions between the center of the pin **204b** and the center of through-hole **222a** are calculated. In addition, similarly, an angle formed by a straight line connecting the center of pair of pins **204a** and the center of pin **204b** and a straight line connecting the center of pair of through-holes **222a** and the center of through-hole **222b** is also calculated.

(49) Similarly, the amount of deviation and the angle of deviation between pins **204** inserted and the through-holes are calculated, and a correction value at the time of inserting a pair of pins into the pair of through-holes is acquired based on the calculated amount of deviation and angle of deviation. By using this correction value, it is possible to insert pair of pins **204a** and **204b** into the centers of two through-holes **222a** and **222b** as illustrated in FIG. **9**, in a case where pair of pins **204a** and **204b** is to be inserted into two through-holes **222a** and **222b**, for example. In other words, pair of pins **204** can be inserted into two through-holes **222** in a state where the amount of deviation of the pins is zero. As a result, for an actual insertion operation, that is, the component mounter, it is possible to improve the insertion accuracy at the time of inserting pair of leads **108** of lead component **106**, of which the component main body has been held by a work head, into pair of through-holes **104** of circuit base material **12** positioned at the work position of the substrate conveyance device.

(50) Note that the imaging of pair of through-holes **222** with pair of pins **204** inserted thereinto, calculation of the amount of deviation and the angle of deviation based on imaging data acquired by the imaging of pair of through-holes **222**, acquisition of a correction value for insertion of the pair of pins into the pair of through-holes that is acquired based on the calculated amount of

deviation and angle of deviation, or the like is performed at the time of inspection of the accuracy of component mounter **10** or at the time of maintenance, inspection, or the like of component mounter **10** after delivery to a user. As a result, it is possible to stabilize the quality of component mounter **10**, avoid problems, maintain the insertion accuracy by periodic maintenance, and the like.

(51) In addition, in component mounter **10**, instead of cut-and-clinch unit **34**, it is also possible to attachably and detachably position and attach, to rotary table **178** of unit moving device **36**, a lead receiving type cut-and-clinch unit receiving a lead component inserted into an insertion hole of a circuit base material. Lead receiving type cut-and-clinch unit **300** is an existing device and when briefly described, as illustrated in FIGS. **12** and **13**, cut-and-clinch unit **300** includes four guide pins **302** that controllably extend from an upper end portion of cut-and-clinch unit **300**.

(52) Four guide pins **302** are disposed to extend in the vertical direction and are arranged at equal pitches on one straight line. Each of four guide pins **302** controllably expands and contracts in length, and is stored in the inside of cut-and-clinch unit **300** in the most contracted state. In addition, a cutting device (not illustrated) for cutting leads of a lead component and a bending device (not illustrated) for bending the leads are stored in the inside of cut-and-clinch unit **300**. With such a structure, in cut-and-clinch unit **300** as well, it is possible to cut and bend leads of a lead component inserted into through-holes **104** of circuit base material **12**.

(53) Specifically, the operation of rotation device **156** is controlled such that the direction of alignment of two through-holes **104** of circuit base material **12** coincides with the direction of alignment of four guide pins **302**. Then, by the operation of X-direction moving device **150** and Y-direction moving device **152**, cut-and-clinch unit **300** is moved below the circuit base material **12** and the positioning thereof is stopped such that the coordinates in the XY-directions of two through-holes **104** of circuit base material **12** coincide with the coordinates in the XY-directions of distal ends of two of four guide pins **302** that are to be inserted into two of the four through-holes. As a result, cut-and-clinch unit **300** is moved along the XY-directions and two through-holes **104** of circuit base material **12** and the distal ends of two of four guide pins **302** are positioned in a state of overlapping with each other in the up-down direction.

(54) Next, cut-and-clinch unit **300** extends two through-holes **104** of the circuit base material and two of the four guide pins **302** that are in a state of overlapping with the two through-holes in the up-down direction. As a result, as illustrated in FIG. **14**, two guide pins **302** are inserted through two through-holes **104** in a direction from a lower side to an upper side and upper ends of two guide pins **302** protrude from the upper surface of circuit base material **12**. In addition, recessed portion **306** is formed on an upper end surface of each guide pin, and the operation of work head moving device **64** is controlled such that a lower end of each of leads **108** of lead component **106** is inserted into recessed portion **306**. That is, the operations of X-direction moving device **68** and Y-direction moving device **70** are controlled such that the coordinates in the XY-directions of the lower ends of pair of leads **108** of lead component **106** held by component holding tool **77** coincide with the coordinates in the XY-directions of the recessed portions of two guide pins **302** exposed at the upper surface of the circuit base material **12**. Then, the operation of Z-direction moving device **72** is controlled such that work heads **60** and **62** are lowered and the lower ends of pair of leads **108** of lead component **106** held by component holding tool **77** are inserted into recessed portions **306** of two guide pins **302** protruding from the upper surface of the circuit base material **12**.

(55) Subsequently, cut-and-clinch unit **300** contracts two guide pins **302** protruding and extending from the upper surface of the circuit base material **12**. At this time, work heads **60** and **62** are also lowered by the operation of Z-direction moving device **72** at the same speed as the speed of contraction of guide pins **302**. As a result, two guide pins **302** contract with the lower ends of pair of leads **108** inserted into recessed portions **306** of two guide pins **302**. Then, as guide pins **302** contract, upper end surfaces of guide pins **302**, that is, recessed portions **306** pass through through-holes **104** of circuit base material **12** and are stored in the inside of cut-and-clinch unit **300**. At this time, the lower ends of pair of leads **108** inserted into recessed portions **306** also pass through pair

of through-holes **104** of circuit base material **12** and are stored in the inside of cut-and-clinch unit **300**. That is, with the lower ends of pair of leads **108** of lead component **106** guided by pair of guide pins **302**, the lower ends of pair of leads **108** are inserted into pair of through-holes **104** of circuit base material **12** and are stored in the inside of cut-and-clinch unit **300**. Then, the lower ends of pair of leads **108** stored in the inside of cut-and-clinch unit **300** are cut and bent by the cutting device. As a result, lead component **106** is mounted on circuit base material **12** in a state where leads **108** are prevented from coming out of through-holes **104**.

(56) As described above, in the case of cut-and-clinch unit **300**, guide pins **302** are extended to be inserted into pair of through-holes **104** from a position below the circuit base material **12** and the distal ends of pair of guide pins **302** protrude from the upper surface of circuit base material **12**. Then, guide pins **302** contract with the lower ends of pair of leads **108** of lead component **106** inserted into recessed portions **306** of the distal ends of guide pins **302**, the lower ends of pair of leads **108** are stored in the inside of cut-and-clinch unit **300**, and the pair of leads are cut and bent. Therefore, although it is desired to suitably insert guide pins **302** into through-holes **104** of circuit base material **12**, a method of improving the accuracy of insertion of guide pins **302** has not been established in a conventional component mounter. Therefore, in the case of component mounter **10**, similarly to the method of improving the accuracy of insertion of leads **108** into through-holes **104** described above, the accuracy of insertion of guide pins **302** is improved by means of imaging data.

(57) Specifically, first, one imaging device **230** is attached to any of sliders **74** and **76** from which a work head has been removed by means of a mechanism for mounting work heads **60** and **62** instead of any of work heads **60** and **62**. As described above, work heads **60** and **62** are positioned and mounted to sliders **74** and **76** respectively such that work heads **60** and **62** can be detached and attached through a one-touch action, and imaging device **230** is positioned and attached to any of sliders **74** and **76** from which work heads **60** and **62** have been removed in a state of facing downward on the vertical line. As described above, imaging device **230** can be attachably and detachably positioned and mounted to rotary table **178** as well and is positioned with high compatibility and attachably and detachably mounted to all of rotary table **178** and sliders **74** and **76**.

(58) Similarly to substrate **220** into which pair of pins **204** of jig **200** are inserted, a substrate into which pair of guide pins **302** of cut-and-clinch unit **300** is inserted is also prepared. Here, the description will be made on the assumption that substrate **220** is used not only for jig **200** but also for guide pins **302** of cut-and-clinch unit **300**. That is, the pitch of two of four through-holes **222** that are adjacent to each other and the pitch of two of four guide pins **302** of the cut-and-clinch unit that are adjacent to each other are the same as each other, four through-holes **222** being formed in substrate **220**. Incidentally, the pitch of two guide pins **302** is a distance between the center of a distal end of first guide pin **302** and the center of a distal end of second guide pin **302**. In addition, the inner diameter of one through-hole **222** is approximately three times the outer diameter of one guide pin **302**.

(59) Substrate **220** having such a structure is conveyed to a work position by conveyance device **50** and is clamped at the work position by clamp device **52**. Then, by imaging device **230** attached to sliders **74** and **76**, through-holes of substrate **220** that are adjacent to each other are imaged from above, and the coordinates in the XY-directions of two of four through-holes **222** that are adjacent to each other are calculated based on imaging data thereof. Meanwhile, in the case of cut-and-clinch unit **300** positioned below substrate **220**, according to the procedure described above, the cut-and-clinch unit is moved and positioned by the operation of unit moving device **36** such that the distal ends of two of four guide pins **302** that are to be inserted into two of the four through-holes of which the coordinates have been calculated coincide with the calculated coordinates in the XY-directions of two through-holes **222** of the substrate that are adjacent to each other. Cut-and-clinch unit **300** extends two guide pins **302**. As a result, two guide pins **302** are inserted into two through-holes **104**, of which the coordinates coincide with the coordinates of two guide pins **302**,

from below. At this time, as illustrated in FIG. 15, cut-and-clinch unit 300 extends guide pins 302 such that the upper end surfaces of guide pins 302 coincide with an upper surface of the substrate 220 in the up-down direction, that is, the upper end surfaces of guide pins 302 and the upper surface of the substrate 220 are flush with each other.

(60) Then, pair of guide pins 302 inserted into two through-holes 222 of substrate 220 positioned within the depth of field of the imaging device is imaged by imaging device 230. In other words, pair of guide pins 302 of the cut-and-clinch unit and two through-holes 222 of the substrate with pair of guide pins 302 inserted therein are simultaneously imaged by one imaging device 230 from above. Note that, as described above, imaging device 230 is a two-dimensional imaging device, and the depth of field of imaging device 230 is relatively narrow. Therefore, the upper ends of guide pins 302 inserted into through-holes 222 are positioned to be flush with the upper surface of substrate 220, so that an appropriate captured image of the upper ends of guide pins 302 and through-holes 222 is acquired.

(61) Then, the amount of deviation between inserted guide pins 302 and through-holes 222 is calculated based on imaging data of two through-holes 222 with pair of guide pins 302 inserted therein. That is, for example, in a case where first guide pin 302a is inserted into first through-hole 222a and second guide pin 302b is inserted into second through-hole 222b as illustrated in FIG. 10, the amount of deviation ($\Delta X1$, $\Delta Y1$) in the XY-directions between the center of first guide pin 302a and the center of first through-hole 222a and the amount of deviation ($\Delta X2$, $\Delta Y2$) in the XY-directions between the center of second guide pin 302b and the center of second through-hole 222b are calculated. In addition, the angle of deviation ($\theta1$) formed by a straight line connecting the center of first guide pin 302a and the center of second guide pin 302b and a straight line connecting the center of first through-hole 222a and the center of second through-hole 222b is also calculated.

(62) Subsequently, through-holes 222 into which two guide pins 302 are inserted are changed and other pair of through-holes 222 is imaged. For example, cut-and-clinch unit 300 is rotated by 90 degrees by the operation of rotation device 156 of unit moving device 36. Then, according to the above-described procedure, two guide pins 302 are inserted into two through-holes 222. At this time, as illustrated in FIG. 11, first guide pin 302a is inserted into first through-hole 222a, and second guide pin 302b is inserted into second through-hole 222c. Then, the amount of deviation ($\Delta X3$, $\Delta Y3$) in the XY-directions between the center of first guide pin 302a and the center of first through-hole 222a and the amount of deviation ($\Delta X4$, $\Delta Y4$) in the XY-directions between the center of second guide pin 302b and the center of second through-hole 222c are calculated. In addition, the angle of deviation ($\theta2$) formed by a straight line connecting the center of first guide pin 302a and the center of second guide pin 302b and a straight line connecting the center of first through-hole 222a and the center of second through-hole 222c is also calculated.

(63) Furthermore, through-holes 222 into which two guide pins 302 are inserted are changed and other pair of through-holes 222 is imaged. For example, cut-and-clinch unit 300 is further rotated by 90 degrees by rotation device 156. Then, according to the above-described procedure, two guide pins 302 are inserted into two through-holes 222. Although not illustrated, first guide pin 302a is inserted into first through-hole 222b, and second guide pin 302b is inserted into second through-hole 222a. Then, the amount of deviation in the XY-directions between the center of first guide pin 302a and the center of first through-hole 222b and the amount of deviation in the XY-directions between the center of second guide pin 302b and the center of second through-hole 222a are calculated. In addition, an angle formed by a straight line connecting the center of first guide pin 302a and the center of second guide pin 302b and a straight line connecting the center of first through-hole 222a and the center of second through-hole 222b is also calculated.

(64) Then, a correction value at the time of inserting pair of guide pins 302 into the pair of through-holes is acquired based on the calculated amount of deviation and angle of deviation. By using this correction value, it is possible to insert two guide pins 302a and 302b into the centers of two through-holes 222a and 222b as illustrated in FIG. 9, in a case where two guide pins 302a and

302b are to be inserted into two through-holes **222a** and **222b**, as illustrated in FIG. 9, for example. In other words, two guide pins **302** can be inserted into two through-holes **222** in a state where the amount of deviation and the angle of deviation are zero. As a result, it is possible to improve the accuracy of insertion at the time of insertion of guide pins **302** into through-holes **104** of circuit base material **12**.

(65) Note that, as with acquisition of a correction value that is performed by using jig **200**, the imaging of pair of through-holes **222** with pair of guide pins **302** inserted therein, calculation of the amount of deviation and the angle of deviation based on imaging data acquired by the imaging of pair of through-holes **222**, acquisition of a correction value for insertion of the pair of guide pins into the pair of through-holes that is acquired based on the calculated amount of deviation and angle of deviation, or the like is performed at the time of inspection of the accuracy of component mounter **10** or at the time of maintenance, inspection, or the like of component mounter **10** after delivery to a user. As a result, it is possible to stabilize the quality of component mounter **10**, avoid problems, maintain the insertion accuracy by periodic maintenance, and the like.

(66) Note that component mounter **10** is an example of a substrate working device. Circuit base material **12** is an example of a substrate. Component mounting device **24** is an example of an inserting device. Clamp device **52** is an example of a holding device. Through-hole **104** is an example of a through-hole. Lead component **106** is an example of a component. Lead **108** is an example of a terminal. Jig **200** is an example of a jig. Component main body **202** is an example of a main body. Pin **204** is an example of a pin. Substrate **220** is an example of a substrate. Through-hole **222** is an example of a through-hole. Imaging device **230** is an example of an imaging device. Cut-and-clinch unit **300** is an example of an inserting device. Guide pin **302** is an example of a pin.

(67) In addition, the present disclosure is not limited to the above-described embodiments, and can be implemented in various forms with various changes and improvements based on the knowledge of those skilled in the art. For example, in the above-described embodiment, pair of pins **204** and two through-holes **222** with pair of pins **204** inserted therein are simultaneously imaged by one imaging device **230** from a lower side of a vertical axis. In addition, pair of guide pins **302** and two through-holes **222** with pair of guide pins **302** inserted therein are simultaneously imaged by one imaging device **230** from an upper side of the vertical axis. Meanwhile, in a case where pair of pins **204** and two through-holes **222** are simultaneously imaged by imaging device **230**, the relative positions of the pair of pins and the pair of through-holes and the angle of alignment, that is, the amount of positional deviation and the angle of deviation can be calculated. In addition, in a case where pair of guide pins **302** and two through-holes **222** are simultaneously imaged by imaging device **230**, the relative positions of the pair of guide pins and the two through-holes and the angle of alignment, that is, the amount of positional deviation and the angle of deviation can be calculated. That is, an image captured by imaging device **230** simultaneously imaging pair of pins **204** and two through-holes **222** with the pair of pins not inserted therein may be acquired, or an image captured by imaging device **230** simultaneously imaging pair of guide pins **302** and two through-holes **222** with the pair of guide pins not inserted therein may be acquired. However, both of pair of pins **204** or the distal ends of guide pins **302** and two through-holes **222** need to be in a state of being able to be simultaneously imaged in a viewpoint from one imaging device **230** that is fixedly disposed and need to be not concealed.

(68) In addition, in the above embodiment, component mounting device **24** for inserting pins **204** into through-holes **222** and imaging device **230** for imaging through-holes **222** with the pins inserted therein are disposed at opposite positions with respect to substrate **220**. In addition, cut-and-clinch unit **300** for inserting guide pins **302** into through-holes **222** and imaging device **230** for imaging through-holes **222** with the guide pins inserted therein are disposed at opposite positions with respect to substrate **220**. Meanwhile, both of component mounting device **24** and imaging device **230** may be disposed above substrate **220** and both of cut-and-clinch unit **300** and imaging device **230** may be disposed below substrate **220**. That is, in a case where pins **204** are inserted

from a position above the substrate **220**, through-holes **222** with pins **204** inserted thereto may be imaged above substrate **220** at an angle offset from the vertical axis. In addition, in a case where guide pins **302** are inserted from a position below the substrate **220**, through-holes **222** with guide pins **302** inserted thereto may be imaged below substrate **220** at an angle offset from the vertical axis.

(69) In addition, although pins **204** of jig **200** extend linearly and vertically from component main body **202** in the above embodiment, pins **204** may be bent or extend from the component main body in a state of being inclined as long as pins **204** are standardized pins, that is, as long as the positions of the pins are known in advance. Although component holding tool **77** grips component main body **202** of jig **200** in the above embodiment, component holding tool **77** may grip a pin. In addition, a jig without a component main body may be adopted. For example, a rod-shaped member may be bent into a U-shape, both ends of the member may be used as substitutes for a pair of pins, and the member may be gripped by a component holding tool and used as a jig. A correction value at the time of acquisition with the component holding tool gripping the pins as described above is used as a correction value used when the component holding tool grips pins of a component and inserts the pins into through-holes of a circuit substrate. In other words, a jig including a pin that is not standardized may be adopted. However, in a case where a jig including a pin that is not standardized is used, it is necessary to image the pin with an imaging device to recognize the position of the pin. In other words, it is possible to use commercially available lead component **106** as jig **200** without using a standardized jig.

(70) In addition, although through-holes **222** are imaged with substantial pins **204** inserted into through-holes **222** in the above embodiment, insubstantial pins (for example, illustrations of pins printed on the jig), holes processed in the jig, or the like may be used instead as substitutes for the pins of the jig. In such a case, for example, illustrations or the like of a linear shape or a rod shape resembling a pin, a lead, or the like may be printed on the component main body of the jig, a pair of holes may be drilled in the component main body of the jig, and the illustrations or the holes and through-holes **222** may be simultaneously imaged for acquisition of a correction value. That is, whether the pins are substantial does not matter as long as shapes, from which the shapes of the pins can be figured out (in other words, shapes of the pins), and images of through-holes **222** are simultaneously captured by imaging device **230**. As described above, jigs of various shapes can be adopted. In addition, various types of component holding tools such as a suction nozzle picking up a main body may be adopted depending on the shape of the jig.

(71) Although the amount of deviation between the center of pin **204** or the guide pin and the center of through-hole **222** is calculated in the above embodiment, various values based on imaging data may be calculated instead of the amount of deviation between the centers as long as the relative positions of pin **204** or the guide pin and through-hole **222** can be figured out. For example, the distance between an inner wall of through-hole **222** and an external line of the lower end surface of pin **204** may be calculated. Note that the same applies to the relative positions of guide pin **302** and through-hole **222**.

(72) Although one piece of two-dimensional image data captured by one two-dimensional imaging device is used in the above embodiment, imaging data captured by another type of imaging device (for example, a stereo camera, a video camera, and a three-dimensional camera) or multiple imaging devices may be used. In addition, imaging data captured by a stereo camera, a video camera, and a three-dimensional camera may be intentionally converted into two-dimensional imaging data and the two-dimensional imaging data may be used.

(73) In addition although an operation of mounting a radial component as lead component **106** has been described in the above embodiment, the present disclosure can also be applied to an axial component. In addition, the present disclosure is not limited to a lead component, and can be applied to a component including a terminal to be inserted into through-hole **222**.

REFERENCE SIGNS LIST

(74) **10**: component mounter (substrate working machine), **12**: circuit base material (substrate), **24**: component mounting device (inserting device), **52**: clamp device (holding device), **104**: through-hole, **106**: lead component (component), **108**: lead (terminal), **200**: jig, **202**: component main body (main body), **204**: pin, **220**: substrate, **222**: through-hole, **230**: imaging device, **300**: cut-and-clinch unit (inserting device), **302**: guide pin (pin)

Claims

1. A substrate working machine comprising: a base material conveyance and holding device configured to convey and hold a substrate in which multiple through-holes are formed, the base material conveyance and holding device including a conveyance device and a clamp device; an component mounting device configured to insert multiple terminals of a component into the multiple through-holes of the substrate held by the base material conveyance and holding device, the component mounting device including a work head and a work head moving device; a jig configured to be held by the component mounting device, the jig including a main body and a pair of pins extending from the main body, a shape of the pair of pins being standardized; and a camera configured to simultaneously image the pair of pins and a pair of through-holes of the multiple through-holes; and a controller including a processor configured to control the component mounting device so that XY-coordinates of the pair of pins of the jig held by the component mounting device coincide with XY-coordinates of first and second through-holes of the multiple through-holes of the substrate, and end surfaces of the pair of pins are flush with a lower surface of the substrate in an up-down direction, and control the camera to acquire a first image data, control the component mounting device so that XY-coordinates of the pair of pins of the jig coincide with XY-coordinates of one of the first or second through-holes and a third through-hole of the multiple through-holes of the substrate, and end surfaces of the pair of pins are flush with a lower surface of the substrate in an up-down direction, and control the camera to acquire a second image data, calculate positions of the first, second, and third through-holes and the shape of the pair of pins based on the acquired first and second image data captured by the camera, calculate translational and angular deviation amounts between centers of the first and second through-holes and centers of the pair of pins, and centers of the one of the first or second through-holes and the third through-hole and centers of the pair of pins, calculate a correction value for inserting multiple terminals of a component into the multiple through-holes of the substrate by the component supply device based on the calculated deviation amounts, and control the component supply device to insert multiple terminals of a component into the multiple through-holes of the substrate based on the correction value.
2. The substrate working machine according to claim 1, wherein the image data is two-dimensional image data.
3. A method comprising: providing the substrate working machine according to claim 1; providing a substrate; controlling the component mounting device so that XY-coordinates of the pair of pins of the jig held by the component mounting device coincide with XY-coordinates of first and second through-holes of the multiple through-holes of the substrate, and end surfaces of the pair of pins are flush with a lower surface of the substrate in an up-down direction, and controlling the camera to acquire a first image data; controlling the component mounting device so that XY-coordinates of the pair of pins of the jig coincide with XY-coordinates of one of the first or second through-holes and a third through-hole of the multiple through-holes of the substrate, and end surfaces of the pair of pins are flush with a lower surface of the substrate in an up-down direction, and controlling the camera to acquire a second image data; calculating positions of the first, second, and third through-holes and the shape of the pair of pins based on the acquired first and second image data captured by the camera; calculate translational and angular deviation amounts between centers of the first and second through-holes and centers of the pair of pins, and centers of the one of the first or

second through-holes and the third through-hole and centers of the pair of pins calculate a correction value for inserting multiple terminals of a component into the multiple through-holes of the substrate by the component supply device based on the calculated deviation amounts, and control the component supply device to insert multiple terminals of a component into the multiple through-holes of the substrate based on the correction value.
